

RODRIGO GOMES DE ARAÚJO

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SUMMARY

MSc Computer Science student focused on systems where **correctness and robustness** are key, such as protocols, consensus and infrastructure other people depend on. Built a **deployable proof-of-work blockchain** in Rust and is currently extending session types to **enforce correct usage of cryptographic resources** at compile-time. Works well **autonomously** and **asynchronously** and emphasises **testing, documentation, and interfacing** in the development process. Comfortable working remote and async.

EDUCATION

Master's <i>Computer Science</i>	Sep 2025 – Present <i>University of Porto (FCUP)</i>
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Bachelor's <i>Computer Science and Engineering</i>	Sep 2022 – Jul 2025 <i>University of Porto (FCUP)</i>
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RESEARCH

CaST (Cryptographically-aware Session Types) <i>Master's Thesis</i>	Jul 2026 – Present <i>FreeST, Haskell, Type Systems, Applied Cryptography</i>
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- Extending the expressiveness of session types to capture **cryptographic resource usage**, so that it's misuse in protocols is detected and rejected at compile-time.
- **Resource tracking composes across nested subprotocols** of arbitrary depth and is preserved under sequential composition and recursion.
- Formalizing the extension and providing a **soundness proof**, with a prototype built on FreeST and applications to smart contracts and protocol specification.

PROJECTS

PoW Blockchain for Decentralized Auctions | Rust, Tokio, LibP2P, Dalek, Docker

- Extensive **research of the state of the art on Bitcoin**, Ethereum and Solana and investigation regarding the security and resilience to failure of these systems.
- Developed a DHT overlay based on Kademlia for **decentralized network construction**.
- PoW Blockchain with **Blake2b hashing, Ed25519 signatures, and Merkle Trees** for transaction validation.
- **Client for auctions** using the blockchain layer as a public ledger.
- Docker **autonomous network behaviour simulation** with clients, bootstrap nodes, well behaved nodes, and malicious nodes.
- Developed for Systems Security class and graded 19.1/20.

Portuguese Elections Analytics Platform | PostgreSQL, PostGIS, Python, Plotly, FastAPI, OCaml

- **Interactive data visualization dashboard** for 10 years of Portuguese elections. Choropleth vote-swing maps, treemaps, radar and multi-year trend charts, drillable from country down to parish.
- **Geospatial rendering in-database**. Maps generated as SVG directly by PostGIS, with topology-preserving simplification adapted to zoom level.
- **ETL pipeline** into a **star-schema warehouse**, feeding **complex analytical queries** that join spatial and electoral data (recursive roll-ups, window functions, OLAP cubes).

CERTIFICATIONS

2nd Place in IEEE RetroJam 2025 IEEE UP Student Branch	Oct 2025
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- Game from scratch in 48 hours, using Rust, learning how to **reduce a problem to its core requirements**.

Academic Merit Awards	2020, 2021, 2022
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Participation in RoboCup 2016 RoboCup Federation	Jun 2016
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1st Place in the National Robotics Championship Instituto Politécnico de Bragança	May 2016
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TECHNICAL SKILLS

Programming Languages: Rust, Haskell, C/C++, OCaml, Java, SQL

Frameworks & Tools: Git, UNIX, Tokio, Docker, Plotly, POSIX, STM, gRPC, LibP2P, Dalek, PostgreSQL

Spoken Languages: Portuguese (Native), English (B2 First, Cambridge), Spanish (Proficient)